

Natural Language Processing

4. Text classification with linear models

Paolo Torrioni

Fall 2025

Notice

Credits

The present slides are largely an adaptation of existing material, including slides by [Jurafsky & Martin](#), [Manning](#), [Potts & MacCartney](#), [Black & Mortensen](#), and others, to whom I am very grateful.

Downloading and sharing

A copy of these slides can be downloaded from [virtuale](#) and stored for personal use only. Please do not redistribute.

Table of Contents

- Text Classification
- Naive Bayes
- Logistic Regression
- Evaluation Methods
- Lexicons for Sentiment, Affect, and Connotation

Text Classification



Sentiment Classification

Text classification or **text categorization**: the task of assigning a label or category to an entire text or document

An example is **sentiment analysis**: the detection of *attitudes*

attitudes: enduring, affectively colored beliefs, dispositions towards objects or persons

- **Holder (source)** of attitude
- **Target (aspect)** of attitude
- **Type** of attitude
 - From a set of types: *Like, love, hate, value, desire, etc.*
 - Or (more commonly) simple weighted polarity: *positive, negative, neutral*, together with *strength*
- **Text** containing the attitude
 - Sentence or entire document



Sentiment Classification

Why sentiment analysis?

Many possible applications! For example:

- **Commercial** (what do people think about new product?)
- **Policy-making** (public sentiment towards public health measure)
- **Prediction** (election outcomes, market trends)
- and more



Sentiment Classification

- Simplest form: binary
- Lexical features may offer informative cues
 - *amazing, great, awesome*
 - *bad, scream, refuse*



Jesus
Chile

Reviewed: 15 September 2020

Great value and a great staff, a real gem

😊 - The location of this place is AMAZING, the staff is great and helpful, there is a nice bar downstairs where I could catch the final of the US open so that was great! The value you get at this place is awesome

😞 - This is manhattan, you never get used to the size of the rooms nor the difficulty finding public parking or paying 50USD overnight parking, but regarding the hotel nothing on their part...



Sergey
United States

Reviewed: 16 June 2020

Bad

😊 - Location is good

😞 - In the next room, a neighbor had a sexual orgy. He screamed all night until 5 in the morning. He was heard by all the neighbors on the floor. We called the reception several times, but no one came to stop him. In the morning we asked to change our rooms, but we were refused this. Only after we said that we would call the police the staff say that they would speak with a neighbor. And finally, he did it when a neighbor started screaming the next night.





Sentiment Classification is Hard

If you are reading this because it is your darling fragrance, please wear it at home exclusively, and tape the windows shut.

(Perfume review in *Perfumes: the Guide*)

She runs the gamut of emotions from A to B

(Dorothy Parker on Katherine Hepburn)

This film should be brilliant. It sounds like a great plot, the actors are first grade, and the supporting cast is good as well, and Stallone is attempting to deliver a good performance. However, it can't hold up.

(Movie review)



Other Text Classification Tasks

• SPAM detection





Other Text Classification Tasks

- **Language id**

- Often the first step in language processing pipelines

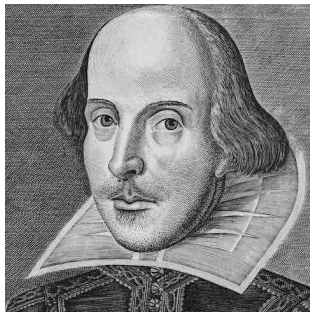




Other Text Classification Tasks

- **Authorship attribution**

- Also: author gender, age, native language
- digital humanities, social sciences, forensic linguistics





Other Text Classification Tasks

- **Subject category classification**

- The reason why Naive Bayes invented in 1961
- Hate speech/misogyny/... /toxic language detection
- Claim/evidence detection, unfair clause detection
- Fake news detection, ...

In fact, many NLP tasks can be seen as classification tasks

- Period disambiguation
- Word tokenization
- Language modeling
- Part-of-speech tagging



Classification Task

Formally:

- Given an input x (often d for document)
- and a fixed set of output classes $Y = y_1, \dots, y_M$ (often c for class)
- return a predicted class $y \in Y$



Approaches to Classification

Rule-based

- Rules may be handwritten
 - blacklist/whitelist
 - regex, e.g., contains \$ and (LOTTERY or "Dear Beloved Friend")
 - Could be highly accurate if carefully refined by expert
 - Could be expensive to build and maintain

In-context learning

- Uses a decoder LLM and a *prompt* describing the task and classes
- May include a few examples (few-shot) or none (zero-shot)

Supervised machine learning

- Uses training set of N labeled documents: $\{(d_i, c_i)\}$
- **Probabilistic** classifier tells probability of observation being in class
 - could be useful piece of information downstream



Supervised Approaches to Classification

Generative approaches

- build a model of how a class could generate some data
- system compares test data point with models to find best fit
- example: naive Bayes

Discriminative approaches

- learn what features are most useful
- without necessarily learning about the classes
- Example: logistic regression

Multinomial Naive Bayes Classifiers

- Representation of a document as a **bag-of-words**

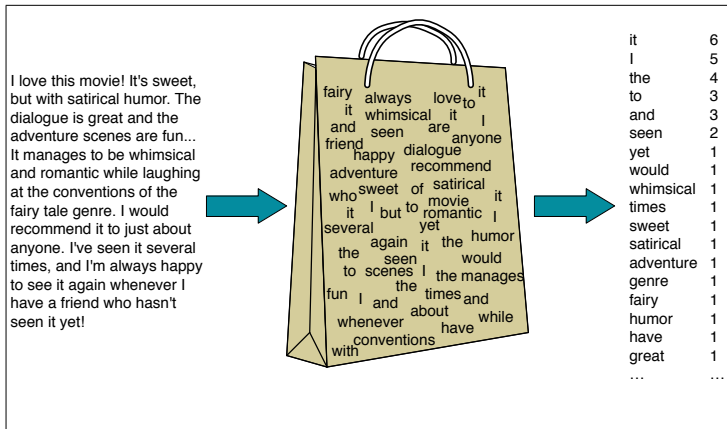


Figure 4.1 Intuition of the multinomial naive Bayes classifier applied to a movie review. The position of the words is ignored (the *bag of words* assumption) and we make use of the frequency of each word.



Multinomial Naive Bayes Classifiers

- Probabilistic classifier
- Given d , it returns the class $\hat{c}$ with max posterior probability

$$\hat{c} = \operatorname{argmax}_{c \in C} P(c|d)$$

- Equivalently,

$$\hat{c} = \operatorname{argmax}_{c \in C} \underbrace{P(d|c)}_{\text{likelihood}} \underbrace{P(c)}_{\text{prior}}$$

- If we represent the document as a set of features $f_1, \dots, f_n$:

$$\hat{c} = \operatorname{argmax}_{c \in C} \underbrace{P(f_1, \dots, f_n|c)}_{\text{likelihood}} \underbrace{P(c)}_{\text{prior}}$$



Multinomial Naive Bayes Classifiers

- Probabilistic classifier

$$\hat{c} = \operatorname{argmax}_{c \in C} \underbrace{P(f_1, \dots, f_n | c)}_{\text{likelihood}} \underbrace{P(c)}_{\text{prior}}$$

- Notice the similarity with **language modeling**
 - the **likelihood** part of the equation can be viewed as a set of class-specific unigram language models
 - each class instantiates a model that assigns a probability to each sentence
 - **generative model**: expresses how to generate the features of a document *if we knew it was of class c*

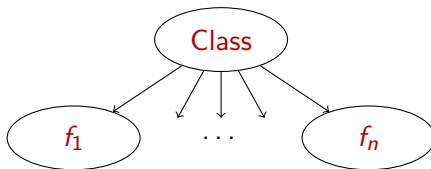


Multinomial Naive Bayes Classifiers

- Probabilistic classifier

$$\hat{c} = \operatorname{argmax}_{c \in C} \underbrace{P(f_1, \dots, f_n | c)}_{\text{likelihood}} \underbrace{P(c)}_{\text{prior}}$$

- Simplifying assumption: mutual feature independence given the class



$$c_{NB} = \operatorname{argmax}_{c \in C} P(c) \prod_i P(f_i | c) = \operatorname{argmax}_{c \in C} P(c) \prod_i P(w_i | c)$$



Multinomial Naive Bayes Classifiers

- In logspace:

$$c_{NB} = \operatorname{argmax}_{c \in C} \log P(c) + \sum_i \log P(w_i | c)$$

(P.S.: why logspace?)

- $P(c)$ and $P(w_i | c)$ learned from frequencies in the data
 - maximum likelihood estimates

$$\hat{P}(c) = \frac{N_c}{N_{doc}}$$

$$\hat{P}(w_i | c) = \frac{\text{Count}(w_i, c)}{\sum_{v \in V} \text{Count}(v, c)}$$



Zero-count and OOV words

$$\hat{P}(w_i|c) = \frac{\text{Count}(w_i, c)}{\sum_{v \in V} \text{Count}(v, c)}$$

- What if w_i does not appear in any document of class c ?

$$\hat{P}_{\text{Laplace}}(w_i|c) = \frac{\text{Count}(w_i, c) + 1}{(\sum_{v \in V} \text{Count}(v, c)) + |V|}$$

- **Laplace smoothing** commonly used in NB categorization
- Ignore words that don't appear at all in the training set
- Stopwords can also be removed from training set
 - Not necessarily performance improvement



Sentiment Classification Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

- $\frac{P(+|predictable\ with\ no\ fun)}{P(-|predictable\ with\ no\ fun)}$
- $P(+)$ $P(-)$
- $P(predictable|+)$ $P(no|+)$ $P(fun|+)$
- $P(predictable|-)$ $P(no|-)$ $P(fun|-)$



Sentiment Classification Example

	Cat	Documents
Training	-	just plain boring
	-	entirely predictable and lacks energy
	-	no surprises and very few laughs
	+	very powerful
	+	the most fun film of the summer
Test	?	predictable with no fun

- $\frac{P(+|predictable\ with\ no\ fun)}{P(-|predictable\ with\ no\ fun)} = \frac{P(+)P(predictable|+)P(no|+)P(fun|+)}{P(-)P(predictable|-)P(no|-)P(fun|-)}$
- $P(+)$ 2/5 $P(-)$ 3/5
- $P(predictable|+) = \frac{0+1}{9+20}$ $P(no|+) = \frac{0+1}{9+20}$ $P(fun|+) = \frac{1+1}{9+20}$
- $P(predictable|-) = \frac{1+1}{14+20}$ $P(no|-) = \frac{1+1}{14+20}$ $P(fun|-) = \frac{0+1}{14+20}$



Optimizing for Sentiment Analysis

① Word occurrence matters more than frequency: **binary NB**

- *fantastic* vs $5 \times$ *fantastic*

		NB Counts		Binary Counts	
		+	-	+	-
Four original documents:					
- it was pathetic the worst part was the boxing scenes	and	2	0	1	0
	boxing	0	1	0	1
	film	1	0	1	0
- no plot twists or great scenes	great	3	1	2	1
+ and satire and great plot twists	it	0	1	0	1
+ great scenes great film	no	0	1	0	1
	or	0	1	0	1
	part	0	1	0	1
	pathetic	0	1	0	1
	plot	1	1	1	1
	satire	1	0	1	0
	scenes	1	2	1	2
	the	0	2	0	1
	twists	1	1	1	1
	was	0	2	0	1
	worst	0	1	0	1
After per-document binarization:					
- it was pathetic the worst part boxing scenes					
- no plot twists or great scenes					
+ and satire great plot twists					
+ great scenes film					

Figure 4.3 An example of binarization for the binary naive Bayes algorithm.



Optimizing for Sentiment Analysis

- ① Word occurrence matters more than frequency: **binary NB**
 - *fantastic* vs $5 \times \textit{fantastic}$
- ② Dealing with **negation**
 - prepend NOT_ to each word between negation and punctuation
 - `didnt like this movie , but I ...`
becomes
 - `didnt NOT_like NOT_this NOT_movie , but I ...`
- ③ Insufficient training data: use specialized **sentiment lexicons**
 - Key resources:
 - **General Inquirer** by Stone et al, 1966
 - **LIWC** by Pennebaker et al, 2007
 - **Hu and Liu's opinion lexicon**, 2004
 - **MPQA subjectivity lexicons** by Wilson et al, 2005
 - Usage: add feature that counts how many words from lexicon occur



Sentiment Datasets

- **IMDB Movie Reviews Dataset** 50K highly polar movie reviews
- **Pang and Lee's Movie Review Datasets** of varying size and labeling schemas
- **Stanford Sentiment Treebank**, 11,855 sentences with fully labeled parse trees for complete analysis of compositional effects of sentiment
- **Multi-Domain Sentiment Dataset** 8K reviews for Amazon products in different categories: books, DVDs, electronics, and kitchen appliances
- **Amazon Product Data** 142.8 million product reviews
- **Semeval** tasks. e.g., in 2020
 - **Task 8: Memotion Analysis**: 8K annotated memes with human-annotated sentiment and humor (sarcastic, humorous, or offensive)
 - **Task 9: Sentiment Analysis for Code-Mixed Social Media Text**: predict the sentiment of a given code-mixed tweet; 39K tweets, focus on the combination of English with Spanish (Spanglish) and Hindi (Hinglish)

Sentiment Datasets

- **Stanford Sentiment Treebank**, 11,855 sentences with fully labeled parse trees for complete analysis of compositional effects of sentiment

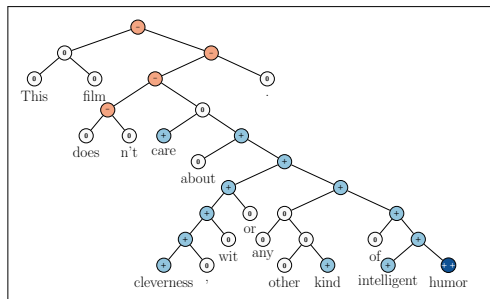


Figure 1: Example of the Recursive Neural Tensor Network accurately predicting 5 sentiment classes, very negative to very positive (−−, −, 0, +, ++), at every node of a parse tree and capturing the negation and its scope in this sentence.



Summary: Naive Bayes is Not So Naive

- Very fast, low storage requirements
- Robust to irrelevant features
 - Irrelevant features cancel each other without affecting results
- However, overly strong independence assumptions
 - Evidence from strongly correlated features is overestimated
- Very good in domains with many equally important features
 - Decision trees suffer from *fragmentation* in such cases – especially if little data
- Optimal if the independence assumptions hold
 - then it is the *Bayes Optimal Classifier* for problem
 - it makes the most probable prediction for a new example
- A **good dependable baseline** for text classification
 - But we will see other classifiers that give better accuracy



Beyond Word Features

- **Byte n-grams**, e.g., for language identification
 - 2-grams ('zw') 3-grams ('nya', ' Vo'), or 4-grams ('ie z', 'thei')
 - can model statistics about word beginning/end
 - a widely used system is [languid.py](#)
- **Feature engineering**, e.g., for spam detection (unsolicited bulk email)
- Predefine likely sets of words and phrases as features and combine with non-linguistic features
 - includes phrase *"one hundred percent guaranteed"*
 - mentions *millions of dollars* (RE)
 - HTML has low ratio of text-to-image area
 - *claims you can be removed from the list*
- **Lexicons** for affective analysis of text, e.g., emotion classification

Logistic Regression



Logistic Regression

- Like Naive Bayes: a **probabilistic classifier** that uses supervised machine learning
- Unlike Naive Bayes: a **discriminative model** that does not attempt to build a model of a class c , but to directly compute $P(c|d)$

Components of a probabilistic ML classifier

- a **feature representation** of the input $x_i^{(j)}$
- a **classification function** computing $\hat{y}$ via $p(y|x)$
- an **objective function** involving minimizing error
- an **optimization algorithm**



Logistic Regression

- input features x_i , weights w_i , bias term b
- weighted sum of evidence for class

$$z = \left(\sum_{i=1}^n w_i x_i \right) + b = w \cdot x + b$$

- **logistic function** to force output to be legal probability

$$P(y = 1) = \sigma(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(w \cdot x + b)}}$$

$$P(y = 0) = 1 - P(y = 1) = \frac{e^{-(w \cdot x + b)}}{1 + e^{-(w \cdot x + b)}}$$

- .5 decision boundary



Example: Sentiment Classification

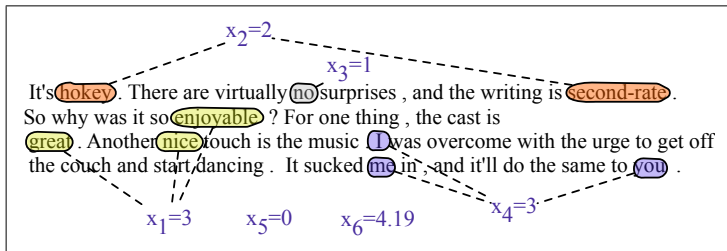


Figure 5.2 A sample mini test document showing the extracted features in the vector x .

• Features:

- x_1 : count(positive lexicon $\in d$)
- x_2 : count(negative lexicon $\in d$)
- x_3 : 1 if "no" $\in d$, 0 otherwise
- x_4 : count(1st, 2nd pronouns $\in d$)
- x_5 : 1 if "!" $\in d$, 0 otherwise
- x_6 : log(word count of d)



Example: Period Disambiguation

- Features:

- x_1 : 1 if "Case(w_i) = Lower", 0 otherwise
- x_2 : 1 if " $w_i \in \text{AcronymDict}$ ", 0 otherwise
- x_3 : 1 if " $w_i = \text{St.} \ \& \ \text{Case}(w_{i-1}) = \text{Cap}$ ", 0 otherwise



General Remarks

- Feature design follows linguistic intuitions
- Possibly large number of features
 - Can use abstract feature specifications (**feature templates**)
 - E.g., a bigram for each pair of words before a period in the training set
- **Sparse** feature space
- Error analysis to provide insights
- For LR and NB, **feature interactions** have to be **designed by hand**
- Recent trend is **representation learning**
- LR robust to correlated features (unlike NB)
- LR works generally better on larger documents or datasets
- NB can work extremely well on small datasets or short documents
- NB fast to train (no optimization step)



Loss Function

- Loss function $L(\hat{y}, y)$: distance between class and prediction
- **Objective**: learn parameters (w, b) that maximize the probability of the correct labels in the training data given the observations $p(y|x)$

$$p(y|x) = \hat{y}^y (1 - \hat{y})^{1-y} = \begin{cases} 1 - \hat{y} & \text{if } y = 0 \\ \hat{y} & \text{if } y = 1 \end{cases}$$

$$\log p(y|x) = y \log \hat{y} + (1 - y) \log(1 - \hat{y})$$

- Hence the **cross-entropy loss** by flipping the sign

$$L_{CE}(\hat{y}, y) = -\log p(y|x) = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$



Optimization Algorithm

- L_{CE} for logistic regression is **convex**
- Procedure to find values of learnable parameters that minimize loss over training set (batch of m examples)

$$\hat{\theta} = \operatorname{argmax}_{\theta} \sum_{i=1}^m \log P(y^{(i)}|x^{(i)}) - \alpha R(\theta)$$

- **Stochastic Gradient Descent** with **mini-batch training**
- **Regularization** $R(\theta)$ to avoid **overfitting**

- **L2 regularization (ridge regression):** $R(\theta) = \|\theta\|_2^2 = \sum_{j=1}^n \theta_j^2$
- **L1 regularization (lasso regression):** $R(\theta) = \|\theta\|_1 = \sum_{j=1}^n |\theta_j|$



Multinomial Logistic Regression

- Sometimes we need more than two classes
- Goal: find the probability of y being in *each* potential class $c \in C$: $p(y = c|x)$
- Extension of LR: **multinomial logistic regression** aka **softmax regression** aka **maxent classifier**
- Uses **softmax** to normalize output into probabilities
- Features are a function of both x and the candidate output class c
- Loss function is sum of logs of $|C|$ output classes

$$L_{CE}(\hat{y}, y) = - \sum_{c=1}^{|C|} 1\{y = c\} \log p(y = c|x)$$

Evaluation Methods



Evaluation Metrics in Binary Classification

Evaluation against **gold labels** or **gold standard**

- Binary classification tasks (detection)
- Contingency table

		<i>gold standard labels</i>		
		gold positive	gold negative	
<i>system output labels</i>	system positive	true positive	false positive	$\text{precision} = \frac{tp}{tp+fp}$
	system negative	false negative	true negative	
		$\text{recall} = \frac{tp}{tp+fn}$		$\text{accuracy} = \frac{tp+tn}{tp+fp+tn+fn}$

Figure 4.4 Contingency table

- Accuracy OK only when classes are balanced
- Most commonly $F_1 = \frac{2PR}{P+R}$



Evaluation Metrics with Multiple Classes

- **Any-of** or **multi-label classification**
- **One-of** or **multinomial classification**
- A separate binary classifier for each class
- Confusion matrix

		<i>gold labels</i>			
		urgent	normal	spam	
<i>system output</i>	urgent	8	10	1	$\text{precision}_u = \frac{8}{8+10+1}$
	normal	5	60	50	$\text{precision}_n = \frac{60}{5+60+50}$
	spam	3	30	200	$\text{precision}_s = \frac{200}{3+30+200}$
		$\text{recall}_u = \frac{8}{8+5+3}$	$\text{recall}_n = \frac{60}{10+60+30}$	$\text{recall}_s = \frac{200}{1+50+200}$	

Figure 4.5 Confusion matrix for a three-class categorization task, showing for each pair of classes (c_1, c_2) , how many documents from c_1 were (in)correctly assigned to c_2



Macroaverage and Microaverage

- **Microaverage:** collect all decisions into single contingency table
 - dominated by most frequent class
- **Macroaverage:** performance per class then average over classes
 - more appropriate when classes equally important

Class 1: Urgent		Class 2: Normal		Class 3: Spam		Pooled									
	true true urgent not		true true normal not		true true spam not		true true yes no								
system urgent	<table> <tr><td>8</td><td>11</td></tr> </table>	8	11	system normal	<table> <tr><td>60</td><td>55</td></tr> </table>	60	55	system spam	<table> <tr><td>200</td><td>33</td></tr> </table>	200	33	system yes	<table> <tr><td>268</td><td>99</td></tr> </table>	268	99
8	11														
60	55														
200	33														
268	99														
system not	<table> <tr><td>8</td><td>340</td></tr> </table>	8	340	system not	<table> <tr><td>40</td><td>212</td></tr> </table>	40	212	system not	<table> <tr><td>51</td><td>83</td></tr> </table>	51	83	system no	<table> <tr><td>99</td><td>635</td></tr> </table>	99	635
8	340														
40	212														
51	83														
99	635														

precision = $\frac{8}{8+11} = .42$

precision = $\frac{60}{60+55} = .52$

precision = $\frac{200}{200+33} = .86$

microaverage precision = $\frac{268}{268+99} = .73$

macroaverage precision = $\frac{.42+.52+.86}{3} = .60$

Figure 4.6 Separate contingency tables for the 3 classes from the previous figure, showing the pooled contingency table and the microaveraged and macroaveraged precision.



Test Sets and Cross-Validation

- To run evaluation with datasets not large enough to be representative: **10-fold cross-validation**
 - randomly choose a training/devset division
 - train classifier
 - compute error rate on held-out test set
 - repeat 10 times then average

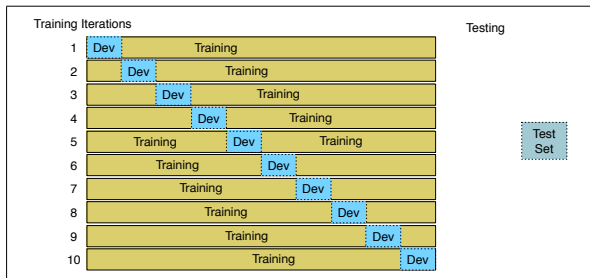


Figure 4.7 10-fold cross-validation



Statistical Significance Testing

Problem: comparing the performance of two classifiers A and B

- Consider a test set $x = x_1, \dots, x_n$ of n observations
- Say A is better than B by $\delta(x)$
- *Is A really better than B? What if A outperformed B on a particular test set **just by chance**? (null hypothesis H_0)*
- If H_0 was true, we would expect A to be better than B by $\delta(x)$ in a number of test sets of size n
 - Given a random variable X over the tests sets,
p-value(x): $P(\delta(X) > \delta(x) | H_0)$, would be expected to be high
- Instead, if p-value(x) was small (less than .05 or .01), we could reject the null hypothesis and agree that A really is better than B
- Standard approaches to computing p-value in NLP are **bootstrap test** and **approximate randomization**



Bootstrapping

Repeated drawing large numbers of smaller samples with replacement

- assume that sample is representative
- obtain multiple virtual test sets
- compute frequency of $\delta(x^{*(i)}) > 2\delta(x)$

	1	2	3	4	5	6	7	8	9	10	A%	B%	$\delta()$
x	AB	AB	AB	AB	AB	AB	AB	AB	AB	AB	.70	.50	.20
$x^{*(1)}$	AB	AB	AB	AB	AB	AB	AB	AB	AB	AB	.60	.60	.00
$x^{*(2)}$	AB	AB	AB	AB	AB	AB	AB	AB	AB	AB	.60	.70	-.10
...													
$x^{*(b)}$													

Figure 4.8 The bootstrap: Examples of b pseudo test sets being created from an initial true test set x . Each pseudo test set is created by sampling $n = 10$ times with replacement; thus an individual sample is a single cell, a document with its gold label and the correct or incorrect performance of classifiers A and B.

Lexicons for Sentiment, Affect, and Connotation



Affective Meaning of Text

Emotion: Relatively brief episode of response to the evaluation of an external or internal event as being of major significance.

(angry, sad, joyful, fearful, ashamed, proud, elated, desperate)

Mood: Diffuse affect state, most pronounced as change in subjective feeling, of low intensity but relatively long duration, often without apparent cause.

(cheerful, gloomy, irritable, listless, depressed, buoyant)

Interpersonal stance: Affective stance taken toward another person in a specific interaction, coloring the interpersonal exchange in that situation.

(distant, cold, warm, supportive, contemptuous, friendly)

Attitude: Relatively enduring, affectively colored beliefs, preferences, and predispositions towards objects or persons.

(liking, loving, hating, valuing, desiring)

Personality traits: Emotionally laden, stable personality dispositions and behavior tendencies, typical for a person.

(nervous, anxious, reckless, morose, hostile, jealous)

Figure 21.1 The Scherer typology of affective states (Scherer, 2000).



Applications

- Deeper linguistic analysis of text
- Closely related to **subjectivity**: study of speaker or writer's evaluations, opinion, emotions, and speculations
- Many applications
 - Linguistic studies (what emotions correlated to religious texts?)
 - Tutorial systems (is student confused, engaged, certain?)
 - **Help lines** (is caller satisfied? trusting? angry? frustrated?)
 - Social media analysis (detect indicators of depression)
 - Meeting summarization (detect stance like friendliness or awkwardness)
 - Conversational agents (personality: extrovert? open to experience?)
- Idea of **lexicons**: instead of using every word as a feature, focus on specific words



What is Emotion?

Two influential theories of emotion

- ① **Basic emotions:** fixed, atomic units, combined into others
 - Elkman 6: *surprise, happiness, anger, fear, disgust, sadness*
 - Plutchik 8: *joy-sadness, anger-fear, trust-disgust, anticipation-surprise*

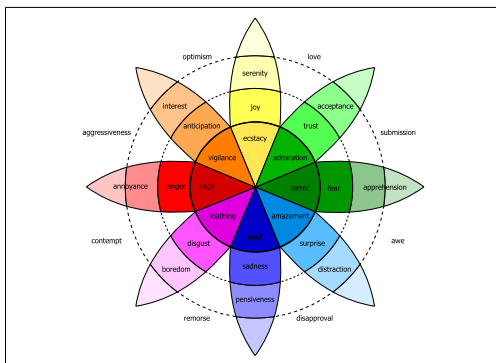


Figure 21.2 Plutchik wheel of emotion.



What is Emotion?

Two influential theories of emotion

- ① **Basic emotions**: fixed, atomic units, combined into others
- ② Emotions described as **continuum** in 2/3-dimensional space
 - **valence**: pleasantness of the stimulus
 - **arousal**: intensity of emotion provoked by stimulus
 - **dominance**: degree of control exerted by stimulus
- **Valence** often used directly as a measure of **sentiment**



Valence of Words

Positive	admire, amazing, assure, celebration, charm, eager, enthusiastic, excellent, fancy, fantastic, frolic, graceful, happy, joy, luck, majesty, mercy, nice, patience, perfect, proud, rejoice, relief, respect, satisfactorily, sensational, super, terrific, thank, vivid, wise, wonderful, zest
Negative	abominable, anger, anxious, bad, catastrophe, cheap, complaint, condescending, deceit, defective, disappointment, embarrass, fake, fear, filthy, fool, guilt, hate, idiot, inflict, lazy, miserable, mourn, nervous, objection, pest, plot, reject, scream, silly, terrible, unfriendly, vile, wicked

Figure 21.3 Some samples of words with consistent sentiment across three sentiment lexicons: the General Inquirer (Stone et al., 1966), the MPQA Subjectivity lexicon (Wilson et al., 2005), and the polarity lexicon of Hu and Liu (2004).



Valence, Arousal, and Dominance of Words

Valence		Arousal		Dominance	
vacation	.840	enraged	.962	powerful	.991
delightful	.918	party	.840	authority	.935
whistle	.653	organized	.337	saxophone	.482
consolation	.408	effortless	.120	discouraged	.0090
torture	.115	napping	.046	weak	.045

Figure 21.4 Samples of the values of selected words on the three emotional dimensions from [Mohammad \(2018a\)](#).

- Some lexicons assign each word a value on all three affective dimensions
- The **National Research Council of Canada (NRC) Valence, Arousal, and Dominance (VAD) Lexicon** assigns scores to 20,000 words



Basic Emotions of Words

Word	anger	anticipation	disgust	fear	joy	sadness	surprise	trust	positive	negative
reward	0	1	0	0	1	0	1	1	1	0
worry	0	1	0	1	0	1	0	0	0	1
tenderness	0	0	0	0	1	0	0	0	1	0
sweetheart	0	1	0	0	1	1	0	1	1	0
suddenly	0	0	0	0	0	0	1	0	0	0
thirst	0	1	0	0	0	1	1	0	0	0
garbage	0	0	1	0	0	0	0	0	0	1

- The **NRC Word-Emotion Association Lexicon** (aka EmoLex) labels 14,000 words using Plutchik's 8 basic emotions



Emotions/Affect Intensity of Words

Anger		Fear		Joy		Sadness	
outraged	0.964	horror	0.923	superb	0.864	sad	0.844
violence	0.742	anguish	0.703	cheered	0.773	guilt	0.750
coup	0.578	pestilence	0.625	rainbow	0.531	unkind	0.547
oust	0.484	stressed	0.531	gesture	0.387	difficulties	0.421
suspicious	0.484	failing	0.531	warms	0.391	beggar	0.422
nurture	0.059	confident	0.094	hardship	.031	sing	0.017

Figure 21.5 Sample emotional intensities for words for anger, fear, joy, and sadness from [Mohammad \(2018b\)](#).

- **NRC Emotion/Affect Intensity Lexicon** contains scores of association for anger, fear, joy, and sadness of 5,814 words



How are Lexicons Created?

- **Expert** annotators
- **Crowdsourcing**, like with **ImageNet**, e.g., using **Mechanical Turk**
 - Each lexicon defines suitable procedure. E.g.:
 - Step 1: answer multiple-choice automatically-generated synonym question (dictionary-based)
 - *Which word is closest in meaning to startle?*
1. automobile 2. shake 3. honesty 4. entertain
 - Step 2: ask annotator to rate association of word with emotion using reference theory (e.g., Plutchik, or Elkman, VAD)
 - *How does startle associate with joy, fear, anger, etc.?*
1. not 2. weakly 3. moderately 4. strongly
- **Semi-supervised** induction of labels
 - Propagate from small set of seed labels
- **Supervised** learning
 - Exploit supervision signals like *online review ratings*

Visualizing Word Sentiment

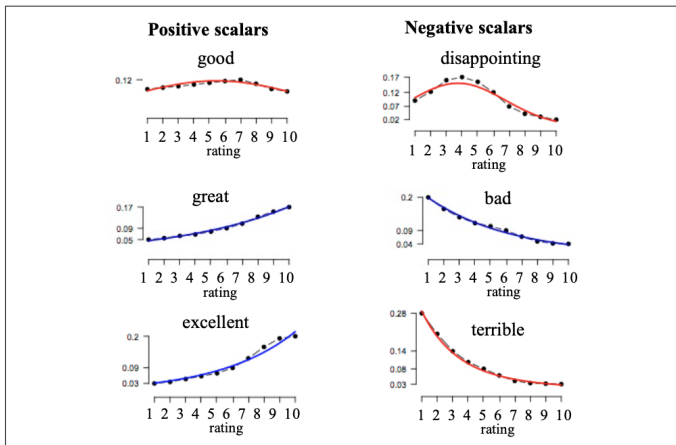


Figure 21.10 Potts diagrams (Potts, 2011) for positive and negative scalar adjectives, showing the J-shape and reverse J-shape for strongly positive and negative adjectives, and the hump-shape for more weakly polarized adjectives.

Visualizing Word Sentiment

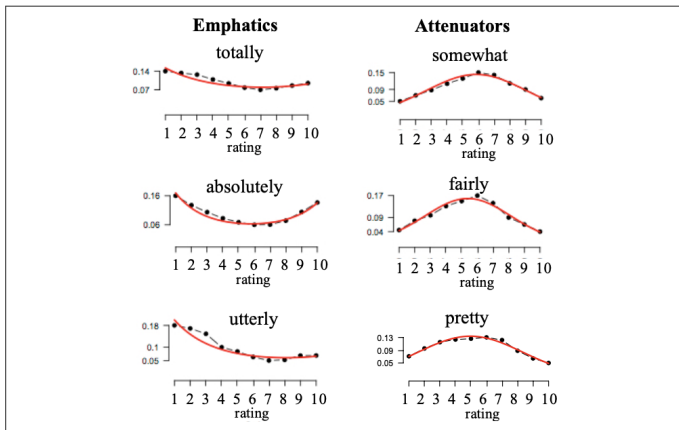


Figure 21.11 Potts diagrams (Potts, 2011) for emphatic and attenuating adverbs.

Questions?